$$(a_1,\ a_2,\ \sqrt{2}a_1a_2,\ \sqrt{2}a_1,\ \sqrt{2}a_2,\ 1)$$

$$(a\ \ b + 1) = \begin{pmatrix} b_1 \\ b_2 \\ \sqrt{2}b_1b_2 \\ \sqrt{2}b_1 \\ \sqrt{2}b_2 \\ 1 \end{pmatrix} = \varphi(a)\ \ \varphi(b)$$

Where: $\varphi(x) = (x_2)$ $\Rightarrow$

$$\begin{pmatrix} x_1 \\ x_2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ 1 \end{pmatrix}$$

$\mathbf{a} = (2)$ $\mathbf{b} = (4)$

$$(1\ ,\ 2\ ,\ \sqrt{2}\ 1 * 2,\ \sqrt{2},\ 2\sqrt{2},\ 1)$$

$$\varphi(a)\ \ \varphi(b) = \begin{pmatrix} 3 \\ 4 \\ \sqrt{2}\ 3 * 4 \\ 3\sqrt{2} \\ 4\sqrt{2} \\ 1 \end{pmatrix} = (1,\ 4,\ 2\sqrt{2},\ \sqrt{2},\ 2\sqrt{2},\ 1) \begin{pmatrix} 9 \\ 16 \\ 12\sqrt{2} \\ 3\sqrt{2} \\ 4\sqrt{2} \\ 1 \end{pmatrix}$$

$$= 9 + 64 + 48 + 6 + 16 + 1 = 144$$

$$|\,a\ \ b + 1\,| = (1 + 3 + 2 * 4 + 1) = 12 = 144$$